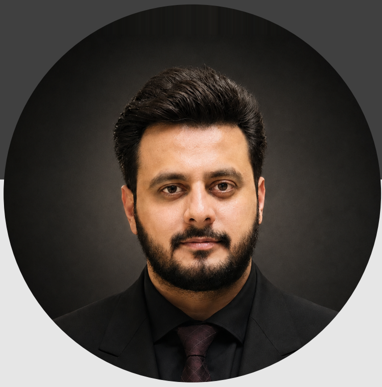
Hafiz Muhammad Ali Zeeshan

Machine Learning | Generative AI | IoT & Emerging Technologies



in/alizeeshan07

Lahore, Pakistan

About me

Advancing next-generation innovation in Agentic AI, LLMs, and Machine Learning; skilled in leading teams to build high performance AI systems, optimize models, and deploy automation that drives business growth.

Areas of Interest

- AI Application Development
- AI Project Management
- Business Automation
- ML, Generative AI, & Agentic AI
- Algorithms & Optimizations
- Deep Reinforcement Learning
- Edge Computing & IoT Networks

AI Tools & Platforms

- LangChain
- Haystack
- Hugging Face
- AWS AI
- Azure AI
- Vertex AI
- LlamaIndex
- CrewAI
- OpenAI API
- Zapier/n8n

Certifications

AWS Certified AI Practitioner
AI/ML, AWS AI services, AWS infrastructure, responsible AI practices. *View*

Databases and SQL for Data Science - IBM
SQL, relational databases, Python integration. *View*

Deep Learning Specialization – Coursera
CNNs, RNNs, Transformers, NLP, transfer learning. *View*

Experience

Jan-2025 – Present
Head of AI | AI Strategy, Business Impact, Automation & Adv. AI Solutions
Adept Tech Solutions Inc. California, USA · Remote
Lead AI strategy, design, and deployment to drive business growth through automation and innovation. Build scalable, high-performance AI systems—multi-agent, MCP, RAG, and fine-tuned models—optimized for cost and speed. Implement cloud, hybrid, edge, and on-device deployments with CI/CD and containerization. Ensure governance, ethics, and compliance, while integrating AI across teams to boost efficiency, reduce costs, and create revenue. Mentor high-performing AI teams and translate technical strategies into business impact.

Nov-2021 – Present
AI Engineer | AI Development, Automation, and SaaS Solutions
Imperium Solutions, USA · Part-time, Remote
Designing, developing, and deploying AI-powered applications including chatbots, automation workflows, and SaaS-based platforms. Expertise in Generative AI (LLMs, Vision, Speech), NLP, Retrieval-Augmented Generation (RAG), multi-agent orchestration, MCP integration, and model fine-tuning. Skilled in inference optimization, cloud-native deployments (AWS, Azure, GCP), MLOps pipelines, and API-first architectures. Experienced in translating business needs into scalable AI solutions, ensuring governance, compliance, and measurable impact.

Dec-2014 – Present
Freelancer | Business Automation, SaaS, Agentic AI & LLM Solutions
Upwork, USA · Remote
AI strategy & solution design, SaaS-based AI applications, Generative AI (LLMs, Vision, Speech), Fine-tuning domain-specific models, NLP and transformers-based systems, Multi-agent orchestration, MCP integration, Retrieval-Augmented Generation (RAG), Inference optimization, AWS cloud deployments, AI-powered process automation, Workflow optimization, Predictive analytics, Anomaly detection, ML/AI research & development, and cross-functional collaboration for business growth.

Nov-2021 – Dec-2024
Principal AI Engineer | Project Management, Generative AI, QC
Adept Tech Solutions Inc. California, USA · Remote
Design and implement advanced AI solutions to support business growth and automation goals. Develop scalable systems—multi-agent, MCP, RAG, and fine-tuned models—optimized for performance and efficiency. Deliver cloud, hybrid, edge, and on-device deployments with CI/CD and containerization. Apply responsible AI practices, ensuring transparency, compliance, and bias mitigation. Collaborate across teams to integrate AI into products and workflows, improving efficiency and enabling new capabilities. Mentor engineers and communicate technical solutions clearly to stakeholders.

Jul-2016 – Nov-2021
Sr. Core Networks Engineer | Troubleshooting, Maintenance & Optimization
Zong CMPak Ltd. Faisalabad, Pakistan · Onsite
Specialized in optimizing telecom core network performance through data-driven analytics, capacity planning, and predictive maintenance. Performed daily maintenance of NSS elements, implemented optimization strategies to improve KPIs, and delivered proactive recommendations for network efficiency. Prepared detailed performance reports, monitored key operational parameters, and resolved customer complaints. Configured and commissioned new equipment, managed subscriber data, and rectified issues related to E1s/PRIs, alarms, and connectivity.

Jan-2015 – Jul-2016
Core Network Engineer | Data Analytics, Network Efficiency & Resilience
Zong CMPak Ltd. Sahiwal, Pakistan · Onsite
Applied data analytics to monitor and maintain traffic channel availability (TCHA) above target thresholds. Resolved service-affecting alarms and optimized radio access network (RAN) performance to ensure stability. Published daily TCHA and alarm tracking reports for proactive maintenance. Managed diesel generator (DG) usage and battery backup (BB) health to reduce operational costs, while addressing external alarms to minimize fuel consumption and improve energy efficiency.

Key AI Projects

Multi-Modal AI Content Creation & Marketing Automation
Built a content pipeline with GPT-4, Claude, Stable Diffusion, and ControlNet, integrated via Zapier, n8n or Make.com, delivering 400% more content at 60% lower cost. *View*

AI-Enhanced Real Estate Investment Platform
QGIS and satellite analysis with automated CMA, forecasting, and MLS integration for 300% faster opportunity detection and 40% higher accuracy. *View*

Intelligent Supply Chain Optimization & Demand Forecasting
LSTM/Transformer models with IoT and Monte Carlo simulations achieving 90%+ forecast accuracy and 35% lower inventory costs. *View*

Computer Vision Quality Control & Inspection System
YOLOv8 and Open3D defect detection (<100ms latency) with PLC integration, delivering 98%+ accuracy at 100+ items/min. *View*

AI-Driven Customer Support & Lead Generation Platform
GPT-4, LLaMA 3, and Whisper API for voice support with BANT qualification, boosting lead conversions by 40%. *View*

Enterprise AI-Powered Document Processing & Workflow Automation
GPT-4, Claude, TrOCR, and CrewAI with FastAPI/Make.com for 95%+ classification accuracy and GDPR/HIPAA compliance. *View*

AI Agents for Marketing Strategy Automation
LangChain and AutoGen agents for automated research, scheduling, and optimization via CrewAI. *View*

Healthcare AI Assistant & Patient Monitoring System
HIPAA-compliant assistant using TrOCR and FHIR, reducing admin work by 70% with real-time vitals monitoring and early alerts. *View*

Soft Skills

- Adaptability
- Critical Thinking
- Research Leadership

Languages

Urdu	●●●●●
English	●●●●○

Degrees

Jan-2024 – Present
Ph.D. EE. (AI & Autonomous Systems) — **CGPA: 4.0/4.0**
SEECS, NUST · Islamabad, Pakistan
Areas of Research: Generative AI, Trustworthy LLMs, Multi-Agent AI Systems, Knowledge Distillation, Quantization, Inference Optimization, Next-Gen IoT & Cyber-Physical Systems, Edge Intelligence, and Intelligent 6G Communication Infrastructures

Sep-2021 – Sep-2023
M.Sc. EE. (AI & AS · With Distinction) — **CGPA: 4.0/4.0**
SEECS, NUST · Islamabad, Pakistan
Majors: Machine Learning (ML), Artificial Neural Networks (ANN), Convex Optimization, Reinforcement Learning, Natural Language Processing, Mobile Robotics, Cloud Computing, and Advanced Wireless Communication

Nov-2010 – Aug-2014
B.Sc. EE. (Electronics & Telecommunications) — **CGPA: 3.34/4.0**
Department of Electrical Engineering, UET · Lahore, Pakistan
Majors: Signals and Systems, Digital Communications, Wireless Communications, Data Structures, Control Systems, Integrated Electronics, Power Electronics and Embedded Systems

Research Collaborations

Sep-2025 – Ongoing
NATO SPS Project – SHAILLA
Science for Peace & Security Programme · Canada, Turkey, Pakistan
Contributing to "Shielding HAPS Against Intelligent Link Level Attacks (SHAILLA)," focusing on AI-driven vulnerability analysis, intelligent attack modeling, LLM-based adversarial strategies, and autonomous countermeasure design. Project aims to evaluate E2E security of next-generation wireless systems using HAPS as base stations, combining agentic AI and ML for robust defense mechanisms.

Jul-2025 – Ongoing
NUST–Huzhou University Joint Research Collaboration
SEECS, NUST - SIE, Huzhou · Pakistan - China
Collaborating with Huzhou University on ML-based wireless communication systems, integrating AI, LLMs, edge computing, and IoT for future-ready 6G networks.

Jan-2024 – Ongoing
Information Proecessing and Transmission (IPT) LAB
SEECS, NUST · Pakistan
Collaborating under the supervision of Dr. Syed Ali Hassan on next-generation wireless systems, focusing on AI, LLMs, autonomous agents, edge computing, and Next-Gen IoT for resilient and intelligent 6G networks.

Publications

- In Progress — **LLM-Enhanced Dynamic Spectrum Management for Integrated Non-Terrestrial and Terrestrial Networks: A Multi-Objective Optimization Approach**, *Mobile Networks and Applications*. Zeeshan, Hafiz Muhammad Ali, et al.
- In Progress — **Towards Intelligent and Sustainable IoT: A DRL Approach for Backscatter-based Communication under QoS Constraints**, *IEEE OJCOMs*. Zeeshan, Hafiz Muhammad Ali, et al.
- May-2025 — **LLM-Based Retrieval-Augmented Generation: A Novel Framework for Resource Optimization in 6G and Beyond Wireless Networks**, *IEEE Communications Magazine COMMAG-24*. Zeeshan, Hafiz Muhammad Ali, et al.
- Apr-2025 — **RAG Powered LLMs for QA: Evolution, Challenges, Applications, and Future Directions**, *ComTech, 2025*. Zeeshan, Hafiz Muhammad Ali, et al.
- Dec-2023 — **DDPG-Based Sum Rate Optimization for Opportunistic Backscatter NOMA Networks**, *IEEE GLOBECOM 2023*. Zeeshan, Hafiz Muhammad Ali, et al.
- Aug-2023 — **MS Thesis: Performance Analysis of CR-NOMA Enabled Backscatter Communications Using Deep Reinforcement Learning**, *NUST*. Hafiz Muhammad Ali Zeeshan.
- Sep-2017 — **Advanced Techniques for Control of Smart Grids**, *IEEE Smart Cities Conference*. Zeeshan, H.M.A., Naeem, M. and Nisar, F.
- May-2017 — **A Review of Distributed Control Techniques for Power Quality Improvement in Micro-Grids**, *IOP Conf. Series: Materials Science and Engineering, Vol. 199, No. 1*. Zeeshan, H.M.A., Nisar, F. and Hassan, A.